The Timing of Early-Life Shocks and Adolescent Mental Health

October 2026

Do the effects of an earthquake experienced at ages 0–4 persist?



Shock 27 February 2010 (27-F), magnitude 8.8

Exposed Children aged 0–4

Effects of early-childhood shocks

Birth weight (Torche, 2011)

Cognitive and non-cognitive development (Gillmore, 2026)

Height and learning (Andrabi, Daniels, and Das, 2023)

Adult mental health (Persson and Rossin-Slater, 2018)

This paper

Mental health in adolescence, ages 14–18

Does it matter at what age children live through the shock?

> "The effects of stress on the brain depend critically on the timing (age of onset and duration). When stress occurs early in life it can have profound and lasting effects on brain organization and function." (Gee and Casey, 2015, p. 184)

Potential mechanisms

	Channels
Socioeconomic and family environment	The earthquake reduces income, damages housing, and increases conflict at home (Conger and Donnellan, 2007; Almond, Currie, and Duque, 2018)
Transmission of maternal stress	Infants regulate their stress through the mother. If she is stressed, she passes it on through daily contact (Persson and Rossin-Slater, 2018)

Data



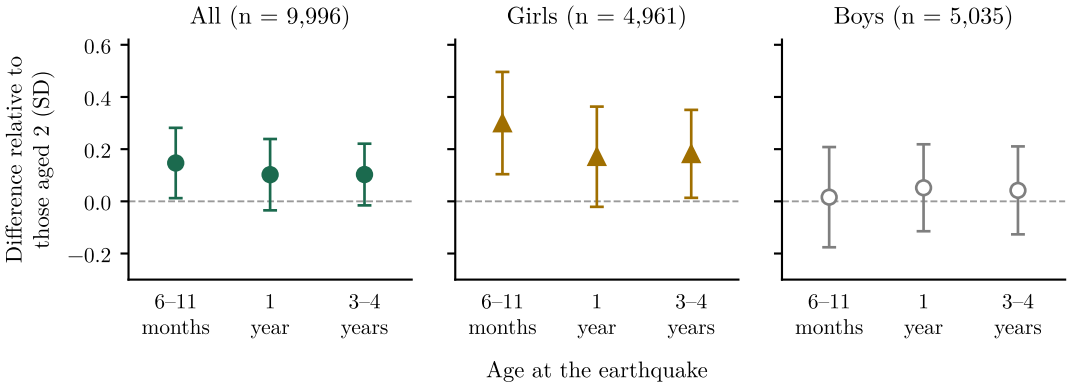
Survey	ELPI, Early Childhood Longitudinal Survey, Chile
Sample	9,996 adolescents born in 2006–2009; 4,961 girls
Respondents	The caregiver in every round; the child in 2017 and 2024

Econometric specification

$$Y_{imc} = \sum_{a \neq 2} \beta_a \left(\text{Age}_{ai} \times \text{Affected}_m \right) + \sum_{a \neq 2} \gamma_a \, \text{Age}_{ai} + \psi_m + \theta_c + X_i' \delta + \varepsilon_{imc}$$

Y_{imc}	Mental health at ages 14–18, PHQ-4
Age_{ai}	Age at the earthquake: 6–11 months, 1 year, or 3–4 years; reference, 2 years
Affected_m	Municipality in the zone of destructive damage
ψ_m, θ_c	Municipality and birth-cohort fixed effects
X_i	Child and household controls

Effect on mental health

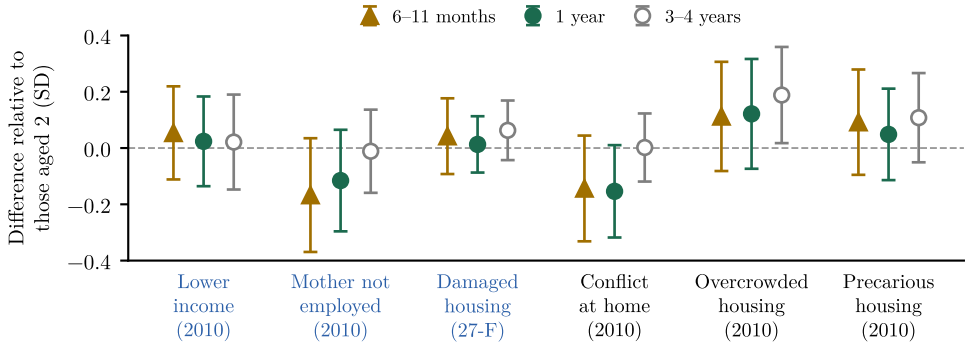


Mental health is measured with the PHQ-4 (Kroenke et al., 2009), four questions answered by the adolescent on symptoms of anxiety and depression (nervousness, worry, loss of interest, and feeling down). Each point is the difference between affected and unaffected municipalities, with 95% CI.

▸ Table A1

Socioeconomic and family environment does not explain the gap

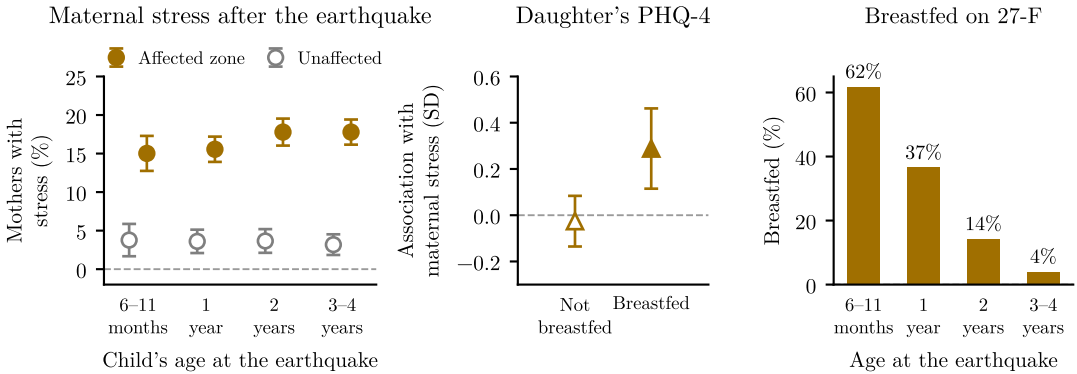
The earthquake worsened the environment (Gillmore, 2026), but with no differences by age



Measured in 2010, one to nine months after the earthquake. In blue, the mechanisms in Gillmore (2026). Lower income, per capita; damaged housing, destroyed or with major damage; conflict at home, the mother shouting, being hostile, or hitting during the visit; overcrowded housing, persons per bedroom; precarious housing, an emergency shack (*mediagua*) or a lent or shared dwelling. Girls, relative to those aged 2; 95% CI.

▸ Table A2

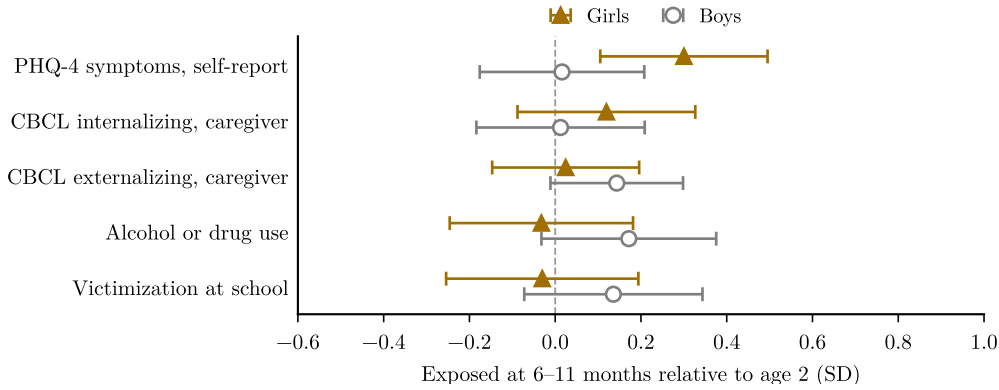
Maternal stress and daughters' mental health by breastfeeding



Active breastfeeding on 27-F serves as a proxy for regulatory dependence and biochemical and behavioral proximity to the mother. Girls, with fixed effects and controls, also for the mother's education and for age; 95% CI.

▸ Table A3 ▸ Table A4

Among boys, only weak signs in behavior and substance use



▸ Table A5 ▸ Table A6

Findings and open questions

What the data suggest

- Girls exposed at 6–11 months have worse mental health than those exposed at age 2, by 0.30 SD on the PHQ-4.
- The socioeconomic and family environment changed similarly at all ages and does not explain this difference.
- Maternal stress is associated with worse mental health only among daughters who were breastfed, 62% of those exposed at 6–11 months.

What remains open

- What explains the effect at ages 3–4, if it operates neither through the socioeconomic and family environment nor through maternal stress?

  Hypothesis: at that age children can already remember consciously what they experience, and the memory of the earthquake could affect their mental health.

References

Almond, Douglas, Janet Currie, and Valentina Duque. 2018. "Childhood Circumstances and Adult Outcomes: Act II." *Journal of Economic Literature* 56 (4): 1360–1446.

Andrabi, Tahir, Benjamin Daniels, and Jishnu Das. 2023. "Human Capital Accumulation and Disasters: Evidence from the Pakistan Earthquake of 2005." *Journal of Human Resources* 58 (4): 1057–1096.

Conger, Rand D., and M. Brent Donnellan. 2007. "An Interactionist Perspective on the Socioeconomic Context of Human Development." *Annual Review of Psychology* 58: 175–199.

Gee, Dylan G., and B. J. Casey. 2015. "The Impact of Developmental Timing for Stress and Recovery." *Neurobiology of Stress* 1: 184–194.

Gillmore, Roberto. 2026. "Natural Disasters and Early Child Development: Evidence from an Earthquake." *Economics of Education Review* 115: 102817.

Kroenke, Kurt, Robert L. Spitzer, Janet B. W. Williams, and Bernd Löwe. 2009. "An Ultra-Brief Screening Scale for Anxiety and Depression: The PHQ-4." *Psychosomatics* 50 (6): 613–621.

Persson, Petra, and Maya Rossin-Slater. 2018. "Family Ruptures, Stress, and the Mental Health of the Next Generation." *American Economic Review* 108 (4–5): 1214–1252.

Torche, Florencia. 2011. "The Effect of Maternal Stress on Birth Outcomes: Exploiting a Natural Experiment." *Demography* 48 (4): 1473–1491.

Table A1. Effect on mental health

	(1) All	(2) Girls	(3) Boys
6–11 months × Affected	0.147**	0.300***	0.016
	(0.069)	(0.100)	(0.098)
1 year × Affected	0.102	0.171*	0.052
	(0.070)	(0.098)	(0.085)
3–4 years × Affected	0.103*	0.182**	0.042
	(0.060)	(0.086)	(0.086)
Observations	9,996	4,961	5,035
R^2	0.066	0.047	0.033
Municipality, cohort, and age-group FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes

The dependent variable is the PHQ-4 score, standardized. Each coefficient is the difference between affected and unaffected municipalities for that age group at the earthquake, relative to those aged 2. WLS with ELPI sampling weights. Controls: age in months, sex, the mother's age and education, household size and rural residence (2010), the mother's number of children, and family history of mental illness (2012). Standard errors clustered by municipality of selection (116 clusters) in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Back

Table A2. The socioeconomic and family environment

	(1) Lower income	(2) Mother not employed	(3) Damaged housing	(4) Conflict at home	(5) Overcrowded housing	(6) Precarious housing
6–11 months × Affected	0.054	−0.167	0.042	−0.144	0.112	0.092
	(0.084)	(0.103)	(0.069)	(0.096)	(0.099)	(0.095)
1 year × Affected	0.024	−0.116	0.013	−0.154*	0.121	0.049
	(0.081)	(0.092)	(0.051)	(0.084)	(0.100)	(0.083)
3–4 years × Affected	0.021	−0.011	0.063	0.002	0.188**	0.108
	(0.086)	(0.075)	(0.054)	(0.062)	(0.087)	(0.081)
Observations	4,479	4,926	4,424	4,702	4,962	4,961
R^2	0.215	0.120	0.102	0.185	0.203	0.089
Municipality, cohort, and age-group FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes

Girls. Each column is an index standardized to mean 0 and SD 1; higher is worse. Measured in 2010, one to nine months after the earthquake; housing damage refers to the earthquake and was reported in 2012. Lower income, per capita; damaged housing, destroyed or with major damage; conflict at home, the mother shouting, being hostile, or hitting during the visit; overcrowded housing, persons per bedroom; precarious housing, an emergency shack or a lent or shared dwelling. Specification as in Table A1. Standard errors clustered by municipality of selection (116 clusters) in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A3. Maternal stress, breastfeeding, and daughters' PHQ-4

	(1) Baseline	(2) Mother's SES	(3) Mother's SES and infancy
Maternal stress × not breastfed	−0.026	−0.005	−0.004
	(0.056)	(0.063)	(0.064)
Maternal stress × breastfed	0.288***	0.280***	0.286**
	(0.089)	(0.096)	(0.112)
Difference, breastfed − not breastfed	0.314***	0.285***	0.290***
	(0.108)	(0.102)	(0.110)
Observations	4,409	4,377	4,377
R^2	0.054	0.053	0.053
Municipality, cohort, and age group × breastfed FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Stress and breastfed × mother's education and employment	No	Yes	Yes
Stress × 6–11 months	No	No	Yes

Girls; PHQ-4 standardized. Maternal stress: the mother reported stress, distress, or anxiety after the earthquake (2012). Breastfed: still breastfeeding on 27-F (2012 report). Column 1 is the estimate in the figure; column 2 lets stress and breastfeeding vary with the mother's education and employment (2010); column 3 also lets stress vary at 6–11 months. Standard errors clustered by municipality of selection (116 clusters) in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Back

Table A4. Maternal stress and breastfeeding by age

	(1) Stress, affected (%)	(2) Stress, unaffected (%)	(3) Breastfed on 27-F (%)
6–11 months	15.0	3.8	61.8
	(1.2)	(1.1)	
1 year	15.6	3.6	36.6
	(0.8)	(0.8)	
2 years	17.8	3.7	14.3
	(0.9)	(0.8)	
3–4 years	17.8	3.2	4.1
	(0.8)	(0.7)	
Children	6,776	2,134	8,880

Means, with standard errors in parentheses, by the child's age at the earthquake. Columns 1 and 2: share of mothers who reported stress, distress, or anxiety after the earthquake (2012), in affected and unaffected municipalities. Column 3: share of children still breastfed on 27-F.

Back

Table A5. Other outcomes, girls

	(1) PHQ-4	(2) CBCL int.	(3) CBCL ext.	(4) Substance use	(5) Victimization
6–11 months × Affected	0.300***	0.119	0.024	−0.032	−0.030
	(0.100)	(0.106)	(0.087)	(0.109)	(0.114)
1 year × Affected	0.171*	0.154	0.100	0.026	0.044
	(0.098)	(0.120)	(0.107)	(0.096)	(0.081)
3–4 years × Affected	0.182**	0.042	−0.082	−0.094	0.081
	(0.086)	(0.095)	(0.083)	(0.083)	(0.092)
Observations	4,961	4,748	4,748	4,961	4,539
R^2	0.047	0.048	0.053	0.115	0.052
Municipality, cohort, and age-group FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Girls. Each column is a separate regression with the specification of Table A1. CBCL internalizing and externalizing scores are reported by the caregiver (2024). Substance use: any alcohol, tobacco, cannabis, or other drug use in the last 12 months. Victimization: peer victimization and class climate. All standardized; higher is worse. Standard errors clustered by municipality of selection (116 clusters) in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Back

Table A6. Other outcomes, boys

	(1) PHQ-4	(2) CBCL int.	(3) CBCL ext.	(4) Substance use	(5) Victimization
6–11 months × Affected	0.016	0.012	0.144*	0.172*	0.136
	(0.098)	(0.100)	(0.079)	(0.104)	(0.106)
1 year × Affected	0.052	−0.063	0.053	−0.064	0.112
	(0.085)	(0.099)	(0.092)	(0.089)	(0.086)
3–4 years × Affected	0.042	0.010	0.126	−0.031	0.098
	(0.086)	(0.099)	(0.090)	(0.079)	(0.094)
Observations	5,035	4,804	4,804	5,034	4,637
R^2	0.033	0.063	0.055	0.110	0.044
Municipality, cohort, and age-group FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Boys. Same outcomes and specification as Table A5. Standard errors clustered by municipality of selection (116 clusters) in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Back